Examiner only

4. *(a)* The table shows information about some sub-atomic particles.

Particle	Symbol	Quark combination	Charge / e	Baryon number
proton	p	uud	+1	1
delta particle	Δ^{++}	uuu	....	....
electron	e^-	no quarks present	....	....
pion	π^-	....	–1	....

(i) **Complete the table.** [3]

(ii) Identify the lepton in the table. .... [1]

(b) JJ Thomson, when studying the properties of cathode rays in 1897, discovered the electron. In the early 20th century, Ernest Rutherford, carrying out a series of experiments on radioactive substances, discovered the proton. The following interaction between protons and electrons has been observed by using high energy particle accelerators.

$$e^- + p \longrightarrow e^- + \Delta^{++} + \pi^-$$

Show how charge and lepton number are conserved in the above interaction. [2]

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(c) The Δ^{++} decays in about 6×10^{-24} s as shown below.

$$\Delta^{++} \longrightarrow p + \pi^{+}$$

(i) Show clearly that both up-quark number and down-quark number are conserved in this decay. [2]

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(ii) Give **two** reasons for believing that this decay is a strong force interaction. [2]

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(d) During a press conference, the spokesman for a nuclear research centre was asked the question:

'You have discovered many new particles, none of which have had any discernible impact on society. How do you justify the huge expense of continuing with these experiments?'

In response, the spokesman referred to the work of JJ Thomson and Ernest Rutherford. Suggest why the spokesman responded in this way. [2]

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